

The empirical recurrence for OEIS A257149, recovered from its tail

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Abstract

OEIS A257149 records “Empirical recurrence of order 79” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 297$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 9$, so the recurrence is proved for every $n \geq 88$. Its last coefficient is $c_{79} = 36028797018963968 \neq 0$, so no further tail satisfies anything shorter; any order-79 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A257149 is “Number of $(n+2) \times (3+2)$ 0..1 arrays with every 3×3 subblock sum of the two medians of the central row and column minus the sum of the minimums of the diagonal and antidiagonal nondecreasing horizontally, vertically and ne-to-sw antidiagonally.”. It has offset 1 and begins

13504, 96680, 732482, 5449304, 39971674, 295114929, 2183417807, 16144616160, ...

The entry states:

Empirical recurrence of order 79 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 15:17:23, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 297$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 297$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 79: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 9$ is the first whose minimal order is exactly the stated 79.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 594$ exact terms from $n = 9$ on, it returns order 79 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{79} c_i a(n-i),$$

c_1	6	c_{21}	−30844993888	c_{41}	−339221853598976	c_{61}	−156032220140666880
c_2	15	c_{22}	−9168313314	c_{42}	−659585144705792	c_{62}	−283765974140715008
c_3	38	c_{23}	102665720975	c_{43}	711594464779776	c_{63}	−168722274612936704
c_4	−128	c_{24}	372825621196	c_{44}	3053637991768576	c_{64}	−7407993146572800
c_5	−2937	c_{25}	487842138459	c_{45}	1128026764234752	c_{65}	300220523072192512
c_6	−2944	c_{26}	−152326720699	c_{46}	−6208136268375040	c_{66}	182064027675918336
c_7	−2578	c_{27}	−1965207824637	c_{47}	−8347675396737024	c_{67}	28324905590194176
c_8	95720	c_{28}	−4778928077332	c_{48}	−5965070567919616	c_{68}	−4573968371548160
c_9	356320	c_{29}	−3893151860759	c_{49}	−2727308383535104	c_{69}	−240790366423351296
c_{10}	381121	c_{30}	5876268874524	c_{50}	19667819546411008	c_{70}	−68815409125523456
c_{11}	−661270	c_{31}	21007992153908	c_{51}	48742967517937664	c_{71}	47776254128029696
c_{12}	−9461518	c_{32}	32118715310440	c_{52}	29824806132187136	c_{72}	61426965864775680
c_{13}	−25347164	c_{33}	12210510098240	c_{53}	−36100483493396480	c_{73}	109979750079922176
c_{14}	−26109074	c_{34}	−59529097840776	c_{54}	−70850808236605440	c_{74}	−18506979718725632
c_{15}	61812155	c_{35}	−131517792437672	c_{55}	−88716521957359616	c_{75}	−15551492463263744
c_{16}	521434773	c_{36}	−81449734285488	c_{56}	−80726877946773504	c_{76}	−67976206875623424
c_{17}	1163832155	c_{37}	85687138167408	c_{57}	97187381737160704	c_{77}	−72620543991349248
c_{18}	856916523	c_{38}	149198801062432	c_{58}	201943663830368256	c_{78}	45035996273704960
c_{19}	−3353769463	c_{39}	202234998605344	c_{59}	119663586346795008	c_{79}	36028797018963968
c_{20}	−17765832414	c_{40}	209353202497152	c_{60}	72189108092403712		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 88$, and it is the recurrence OEIS A257149 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 9$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{79} - c_1x^{79-1} - \dots - c_{79}$. Its constant term is $-c_{79} = -36028797018963968$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 79 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 79, so $\deg q = 79$, and it is monic. It holds from some index t on. From $\max(t, 9)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 79, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 79, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 36028797018963968.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-79 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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